

DEVELOPMENT OF AN AI-POWERED RAG CHATBOT FOR INTELLIGENT BUSINESS KNOWLEDGE RETRIEVAL

INTERNSHIP REPORT

Submitted by

**SHIVA HARINY MR
25PDS506**

In partial fulfilment of the requirement
for the award of the degree of
MASTER OF SCIENCE IN DATA SCIENCE

Under the guidance of
Dr. Y. SUNIL RAJ, M.Sc., M.Phil., Ph.D.
Assistant Professor of Data Science



**DEPARTMENT OF DATA SCIENCE
ST. XAVIER'S COLLEGE (AUTONOMOUS)**

(Recognized as 'College with Potential for Excellence' by UGC)
(Accredited by NAAC at A⁺⁺ Grade with a CGPA of 3.66 out of 4 in IV Cycle)

PALAYAMKOTTAI – 627002

July – 2026

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PALAYAMKOTTAI -627002



BONAFIDE CERTIFICATE

This is to certify that the internship work entitled “**DEVELOPMENT OF AN AI-POWERED RAG CHATBOT FOR INTELLIGENT BUSINESS KNOWLEDGE RETRIEVAL**” is the bonafide work of **SHIVA HARINY M R (25PDS506)** who carried out the internship under my supervision and submitted during the academic year 2026 – 2027.

GUIDE

HEAD OF THE DEPARTMENT

The Viva-Voce held on

INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I **SHIVA HARINY M R**, a student enrolled in Department of Data Science, at St. Xavier's College (Autonomous), Palayamkottai, hereby declare that the documentation presented for my internship, titled "**DEVELOPMENT OF AN AI-POWERED RAG CHATBOT FOR INTELLIGENT BUSINESS KNOWLEDGE RETRIEVAL**", is entirely my original work. I assert that this has been done under the supervision and guidance of **Dr. Y. SUNIL RAJ, M.Sc., M.Phil., Ph.D.** and in accordance with the rules and regulations set forth by Department of Data Science for the completion of the internship.

Place :

Date :

Student's Signature

Date: 15/05/2026

To: SHIVA HARINY M R

Register Number: 25PDS506

College Name: ST. XAVIER'S COLLEGE PALAYAMKOTTAI (AUTONOMOUS)

Subject: Offer for Short-Term Internship

We are pleased to offer you a 30-day educational internship with **Inavate** ("the Company"). During this term, you will be assigned as a **Developer - Intern**, specifically supporting the development project for our client, **Velantri Labs**.

Term and Duration

This is a fixed-term internship for a period of **30 days**, commencing on **15/05/2026** and concluding on **15/06/2026**. This internship will end automatically on the conclusion date without further notice.

Role and Learning Objectives

In this role, you will work under the guidance of our technical leads to assist in the development and optimization of digital products.

Your core learning areas will include:

- Assisting in the development of functional modules and translating technical requirements into clean, modular, and efficient code.
- Identifying logic gaps and debugging issues within assigned tasks to ensure smooth integration with the broader system.
- Collaborating on the creation and maintenance of technical records, including code comments and progress logs, to ensure transparency in the development process.
- Participating in the testing and validation of your code to meet the performance and quality standards of the project.

Unpaid Nature of Internship

This is an **unpaid internship** designed for your professional development and academic exposure. You will not be eligible for a stipend, salary, or any other monetary compensation or company-sponsored benefits during this period.

Nature of Engagement

This internship is for the purpose of skill acquisition and project-based learning. While Inavate

regularly reviews intern performance for future opportunities, this offer is for a fixed term and does not guarantee, nor imply, any entitlement to a permanent position at Inavate or its clients upon completion.

Confidentiality

In the course of your internship, you will have access to proprietary information belonging to both Inavate and Velantri Labs. You agree to maintain absolute confidentiality and will not disclose any data, strategies, or internal documents to any third party during or after your term.

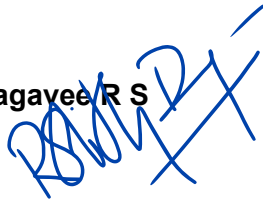
Termination

Either party may terminate this internship at any time by providing **2 days** written notice.

Certification

Upon the successful completion of the internship and the satisfactory handover of all project-related tasks, you will be issued an **Internship Completion Certificate** by Inavate.

For Inavate: **Vithya Ragavee R S**



Accepted by:



SHIVA HARINY M R



INTERNSHIP CERTIFICATE

FOR COLLEGE STUDENTS

This is to certify that

Shiva Hariny M R

of ***St. Xavier's College, Palayamkottai (Autonomous)***

has successfully completed an internship program from

15/05/2026

to

15/06/2026

in the role of ***Developer - Intern***

Roles & Responsibilities:

Backend Development, Frontend Development,

RAG Pipeline & Retrieval Engineering, API & Database Integrations,

AI Observability (Langfuse) & Guardrails

This certificate recognizes the dedication and valuable contributions made during this internship. We believe the experience gained here will be a significant asset to their future career.

Inavate 325, Narayana Nagar, Sankar Nagar,
Address: Tirunelveli, Tamil Nadu 527357
Contact: +91 7259285000

A blue ink handwritten signature, appearing to read 'Rishabh', written over a horizontal line.

Authorized Signature

Date	Intern Name	Task	Planned Work for Today	Actual Work Done	Time Spent (hrs)	Difficulty Level	Blockers / Problems Faced	Support Needed	Status	Learnings / Notes
18 May	Shiva Hariny M R	Dataset and Schema task	Creating sample datas and familiarizing with the schemas	Created an extended seed file containing 25 new entries and 3 txt sample files. Successfully installed PostgreSQL, configured pgvector, executed the seed SQL file, and verified the database schema and tables.	6 hrs	Medium	Struggled with PostgreSQL and pgvector installation, PATH configuration issues, and resolving duplicate/extension-related SQL errors during setup.	Guidance for pgvector setup and database verification	Completed	Learned PostgreSQL setup, pgvector installation, database schema structure, SQL seed execution, table relationships, and basics of vector database setup for RAG architecture.
21 May	Shiva Hariny M R	Extraction & Langfuse Setup	Implement Lead extractor and Order extractor using PostgreSQL and SQLAlchemy, generate human-readable outputs, and integrate Langfuse tracing	Configured PostgreSQL database connection using SQLAlchemy and dotenv, implemented extract_leads.py and extract_orders.py, handled NULL values and FK relationships, generated readable extraction outputs (45 leads, 11 orders), integrated Langfuse tracing successfully, created output log files, updated, gitignore, cleaned GitHub branch structure and pushed code to a separate branch	8 hrs	Hard	Faced database connection issues due to special characters in DB password, Unicode encoding issues while exporting output files, GitHub branch and gitignore cleanup issues, and Langfuse SDK version/API compatibility problems	Guidance for GitHub branch cleanup, .gitignore handling, Langfuse integration, and debugging SDK/API errors	Completed	Learned SQLAlchemy DB integration, environment variable handling with dotenv, NULL-safe extraction formatting, Git cleanup using gitignore and git rm --cached, Langfuse observability setup, and professional GitHub branching workflow for internship tasks
23 May	Shiva Hariny M R	RAG Pipeline & Validation Testing	Implement retrieval pipeline, cosine similarity retrieval, LangChain + OpenAI integration, Langfuse tracing, validation testing and report generation	Built retrieval function, implemented cosine similarity search, integrated OpenAI + LangChain + Langfuse, added interactive PowerShell question input, performed Retrieval Precision Test, Threshold Filtering Test, Empty Context Test, Retrieval Consistency Test, Token-Latency Verification, created validation report and regression tested complete RAG workflow	6-7 hrs	Medium	OpenAI API configuration issues, Langfuse SDK compatibility issues, retrieval consistency tuning, token verification SDK issue	Debugging support for LangChain/OpenAI integration and validation testing approach	Completed	Learned semantic retrieval, cosine similarity retrieval, RAG workflow, guardrails against hallucination, Langfuse observability, validation engineering, regression testing and production stability improvements
24 May	Shiva Hariny M R	Session Tracing + CSV Parser + RAG Stability Improvement	Implement Langfuse session tracing, build CSV ingestion pipeline, improve RAG retrieval stability and validate production workflow	Added session-level tracing in Langfuse using UUID session IDs, grouped traces under sessions and validated dashboard observability. Built CSV parser pipeline with metadata generation, batch processing, missing value handling and integration testing. Improved RAG retrieval consistency by threshold tuning, query normalization improvements, increasing retrieval top_k, reranking logic addition and retrieval validation testing. Verified end-to-end workflow stability and production readiness.	7-8 hrs	Medium	Langfuse session integration understanding, retrieval inconsistency, threshold tuning, ranking instability, semantic retrieval mismatch, parser validation challenges	Guidance for Langfuse session tracing, retrieval optimization strategy and validation approach	Completed	Learned session-based observability, Langfuse tracing architecture, ingestion pipeline design, metadata generation, retrieval optimization, reranking concepts, threshold filtering, semantic search tuning, batch processing and production-grade RAG stability improvement techniques
26 May	Shiva Hariny M R	Frontend Development for Worklyft RAG Intelligence	Build React frontend for Worklyft RAG Intelligence platform with PDF and CSV upload capability, integrate backend APIs, create user-friendly UI	Set up React frontend environment, installed required dependencies, created frontend structure, integrated upload functionality planning for PDF- and CSV files, configured API integration paths, fixed Python import resolution issues (csv_parser, pdf_parser), configured VS Code/PyLance settings, verified backend API execution, removed UI favicon issue	7 hrs	Hard	Import resolution issues in VS Code (csv_parser, pdf_parser), frontend npm script confusion (npm start vs npm run dev), PyLance warnings, favicon/browser cache issue	Guidance for Python path configuration and frontend setup	Completed	Learned React frontend setup, Vite workflow, API integration concepts, Python module path management, VS Code environment configuration, frontend-backend integration flow, and demo presentation structure
29 May	Shiva Hariny M R	Demo Feedback Fixes & System Stability Improvements	Fix bugs and improvements identified during demo discussion, improve retrieval consistency, and enhance system reliability	Performed bug fixing based on demo feedback, improved RAG retrieval consistency, refined query handling behavior, fixed retrieval instability issues, validated customer isolation flow, improved system response reliability, verified frontend-backend flow, and tested stability after fixes	3 hrs	Medium	Retrieval inconsistency in similar query variations, response reliability issues, minor integration fixes identified during demo	Guidance on retrieval improvements and system stability expectations	Completed	Learned retrieval optimization techniques, production debugging workflow, customer isolation validation, and importance of post-demo iterative improvements

Date	Intern Name	Task	Planned Work for Today	Actual Work Done	Time Spent (hrs)	Difficulty Level	Blockers / Problems Faced	Support Needed	Status	Learnings / Notes
30 May	Shiva Hariny M R	Task 6 — Guardrail Layer Implementation & Validation	Implement guardrail layers for RAG system, improve customer-scope retrieval, add validation pipeline, and prepare test documentation	Implemented Layer 1 similarity threshold filtering. Layer 2 prompt hardening flow. Layer 3 response validation, integrated guardrail pipeline into RAG workflow, added prompt injection protection, validated customer isolation behavior, tested rejection of unrelated queries, verified source citation support, generated guardrail_test_results.mtd, and performed latency and retrieval verification tests	8 hrs	Hard	Threshold tuning, retrieval validation balancing, ensuring unrelated queries are blocked without affecting valid customer retrieval	Guidance on guardrail expectations and testing scenarios	Completed	Learned production guardrail design, similarity threshold filtering, prompt hardening, response validation pipeline, prompt injection prevention, retrieval validation strategy, and production-level RAG safety architecture
				Implemented API improvements for /chat endpoint, added request validation handling, integrated mandatory session_id flow, improved customer validation checks, verified source citation response structure, validated customer isolation behavior, tested FastAPI endpoint responses using Swagger and curl, improved response formatting, verified health endpoint behavior, and performed merge conflict resolution while synchronizing branches						
1 June	Shiva Hariny M R	Task 7 (A) – API Layer Completion & Validation	Complete API layer requirements, improve request validation, verify endpoint functionality, and finalize source citation handling	Designed architecture for domain-restricted external retrieval integration, planned hybrid retrieval flow combining internal RAG + approved external business-domain sources, defined domain validation logic, designed external retrieval guardrails to prevent general knowledge answering, prepared source whitelisting approach, designed business-domain intent classification layer, planned semantic domain matching flow, documented domain citation requirements, and finalized technical architecture for controlled external knowledge integration without affecting customer isolation or existing guardrails	8 hrs	Hard	Merge conflicts during branch synchronization, API contract mismatches, validation flow adjustments, source response structure alignment	Guidance on API contract expectations and merge conflict verification	Completed	Learned API contract enforcement, session trace propagation, structured validation handling, customer isolation verification, merge conflict resolution workflow, and production API testing methodology
3 June	Shiva Hariny M R	External Source Citation Architecture Design & Domain Restricted Retrieval Enhancement	Design external source citation support while maintaining strict domain restrictions and preventing generic knowledge access	Researched SerpAPI and analyzed its integration capabilities for enterprise RAG systems. Explored how external retrieval can be restricted to customer/business-related entities instead of allowing generic internet queries. Created and tested sample business datasets for external retrieval scenarios such as CEO information, market data, and company-related updates. Configured Google Custom Search and compared it with SerpAPI approaches. Updated branches with latest upstream changes using Git merge workflow and resolved synchronization-related issues for PR review readiness.	7 hrs	Hard	Designing external retrieval safely without enabling unrestricted GPT behavior, maintaining domain-only access restrictions, preventing general knowledge leakage	Guidance on external retrieval scope validation and business domain restrictions	Completed	Learned hybrid retrieval architecture concepts, domain restricted retrieval design, external source governance principles, intent classification strategies, semantic domain matching concepts, and production guardrail expansion planning
28 May 2026	Shiva Hariny M R	Task 7 (B) – External Source Retrieval	Planned to research and integrate external source retrieval for domain-restricted RAG responses, test SerpAPI capabilities, and synchronize the branch with the latest upstream changes before PR review.	Created comprehensive documentation for Source Confidence Score and Smart Duplicate Detection. Defined business use cases, proposed confidence score calculation logic, designed duplicate detection workflow using embeddings and cosine similarity, outlined FastAPI endpoints, and documented external source citation strategy for chatbot responses.	7 hrs	Hard	Faced issues understanding external retrieval architecture, Google Custom Search limitations, and branch synchronization workflow during merge operations.	Needed clarification regarding enterprise-level external retrieval guardrails and safe domain-restricted internet search logic.	Completed	Learned the architectural difference between internal RAG retrieval and controlled external retrieval. Understood how enterprise AI systems safely allow external searches only for customer/domain-related entities and prevent generic question answering. Also gained practical experience in Git branch synchronization and merge workflows.
29-May-2026	Shiva Hariny M R	RAG Chatbot Feature Documentation & Design	Prepare detailed documentation for Source Confidence Score and Smart Duplicate Detection features, including use cases, tech stack, logic, formulas, and API design.		8 hrs	Medium	Finalizing confidence scoring factors and duplicate similarity threshold values for implementation.	Clarification on production-level scoring weights and similarity threshold recommendations.	Completed	Learned how confidence scoring improves trust in AI-generated responses, how semantic similarity can be used for duplicate detection, and the importance of source citation for transparency in RAG-based systems.

Date	Intern Name	Task	Planned Work for Today	Actual Work Done	Time Spent (hrs)	Difficulty Level	Blockers / Problems Faced	Support Needed	Status	Learnings / Notes
30-May-2026	Shiva Hariny M R	External Source Citation & Guardrail Improvements	Improve external source retrieval and citation display	Implemented external source citation feature, improved entity validation, fixed external retrieval routing issues, tested AWS, CrowdStrike, Azure and other domain-specific queries	8 hrs	Hard	Inconsistent external retrieval and unrelated entity matching	Guidance on retrieval validation approach	Completed	Learned how guardrails, entity validation and external retrieval interact in RAG systems
31-May-2026	Shiva Hariny M R	Multi-Turn Memory Integration	Integrate session-based conversation memory into chatbot	Merged session memory module, updated API integration, improved reference resolution for pronouns (he, him, his, that company, etc.), tested multi-turn conversations across customers	8 hrs	Hard	Memory references were not resolving correctly across conversations	Support in identifying reference resolution edge cases	Completed	Learned session management, conversation state handling, and context propagation
01-Jun-2026	Shiva Hariny M R	Reporting Queries & Structured Retrieval	Improve reporting queries and reduce hallucinated responses	Implemented reporting query pipeline for leads, orders, strategies, completed active/lost/won status queries, added grounding validation, fixed retrieval contamination issues, tested customer-specific reporting	9 hrs	Hard	Reporting queries were mixing summary chunks and record chunks	Clarification on structured retrieval design	Completed	Learned importance of structured retrieval over similarity search for reporting-style questions
02-Jun-2026	Shiva Hariny M R	Source Confidence Score & PR Review Fixes	Implement confidence scoring and resolve code review comments	Developed Source Confidence Score feature, integrated confidence calculation into API and frontend, fixed confidence propagation issues, resolved pytest failures, addressed mentor PR review comments, improved code quality and testing coverage	9 hrs	Hard	Confidence values not propagating correctly between backend and frontend	Review feedback and testing guidance	Completed	Learned confidence scoring, metadata-based source validation, frontend-backend integration, and production-quality testing practices
03-Jun-2026	Shiva Hariny M R	Retrieval Pipeline Analysis	Analyze document retrieval flow and source selection logic	Traced retrieval pipeline, source ranking, uploaded file prioritization, entity resolution, and external retrieval interactions. Investigated source contamination issues in document QA.	8 hrs	Medium	Difficult to identify where unrelated documents entered retrieval results.	Guidance on retrieval architecture.	Completed	Learned end-to-end retrieval flow, source scoring, and document ranking mechanisms.
04-Jun-2026	Shiva Hariny M R	Version Tracking Enhancement	Study current version tracking implementation and identify gaps	Analyzed file_registry version chains, previous_id relationships, latest version selection logic, and retrieval restrictions. Designed version-aware retrieval approach and history queries.	7 hrs	Medium	Understanding how historical versions are hidden from normal retrieval.	Discussion on version architecture.	Completed	Learned document versioning concepts and retrieval isolation strategies.
05-Jun-2026	Shiva Hariny M R	Version Comparison Feature	Implement version history and comparison functionality	Developed version history queries, specific version retrieval, latest/previous version access, version counting, timestamp display, and comparison workflows. Added regression testing for version operations.	9 hrs	Hard	Comparison output initially showed raw content instead of meaningful differences.	Review of comparison logic.	Completed	Learned version comparison design and document diff workflows.
06-Jun-2026	Shiva Hariny M R	Diff Engine Improvement	Improve comparison quality and change detection	Investigated comparison failures. Identified chunk-format limitations. Modified comparison engine to analyze original document structure. Added field-level change detection, section grouping, added/removed content tracking and timeline reporting.	9 hrs	Hard	Raw chunk data lost document structure making comparison difficult.	Technical discussion on diff generation.	Completed	Learned structured document parsing and generic diff generation techniques.
07-Jun-2026	Shiva Hariny M R	Query Intent Detection Fixes	Improve handling of version-related user questions	Enhanced detection for comparison, additions, removals, sections, timestamps, history, and upload-date queries. Added topic filtering and focus-based comparison responses. Fixed edge cases discovered during testing.	8 hrs	Hard	Many user query variations were not triggering comparison logic correctly.	Validation of intent patterns.	Completed	Learned intent classification and natural-language query handling.
08-Jun-2026	Shiva Hariny M R	Bug Resolution	Validate demo scenarios and address mentor feedback	Conducted extensive testing of CRM retrieval, reporting queries, version history, version comparison, gold-rate comparison, and document retrieval. Investigated confidence scoring, source labeling, uploaded-file restrictions, and CSV version-tracking scenarios. Prepared technical explanations for mentor review.	8 hrs	Medium	Multiple edge cases discovered during demo testing.	Mentor feedback and review.	Completed	Learned practical debugging, demo preparation, and explaining system architecture to stakeholders.

Date	Intern Name	Task	Planned Work for Today	Actual Work Done	Time Spent (hrs)	Difficulty Level	Blockers / Problems Faced	Support Needed	Status	Learnings / Notes
09-06-2026	Shiva Hariny M R	Excel Version Tracking Research	Design a mechanism to detect additions, deletions and modifications in uploaded Excel datasets.	Implemented change detection for uploaded datasets to identify row additions, deletions, and modifications. Developed a comparison engine to detect inserted, updated, and deleted records between CSV uploads.	6 hrs	Hard	Selecting an efficient approach for detecting edits without reprocessing the complete dataset.	Discussion with mentor regarding version tracking strategy.	Completed	Learned techniques for dataset versioning, row comparison and change detection.
10-06-2026	Shiva Hariny M R	Retrieval Pipeline Enhancement	Improve retrieval flow and define multiple knowledge sources	Enhanced retrieval flow and integrated multiple knowledge sources.	8 hrs	Hard	Deciding routing logic between different retrieval sources.	Architecture review with mentor.	Completed	Understood hybrid retrieval design and source routing concepts.
11-06-2026	Shiva Hariny M R	Platform Architecture Planning	Understand future expansion of the chatbot platform.	Researched and documented future expansion requirements.	9 hrs	Hard	Designing reusable architecture while keeping existing system intact.	Mentor guidance on long-term architecture.	Completed	Learned platformization concepts, API-first design and reusable service architecture.
12-06-2026	Shiva Hariny M R	Product Architecture & API Design	Design scalable chatbot architecture.	Designed scalable chatbot architecture	5 hrs	Medium	Understanding modular architecture for future integrations.	Design discussions with mentor.	Completed	Learned service-oriented architecture and reusable chatbot engine design.
13-06-2026	Shiva Hariny M R	Visualization & KPI Planning	Improve visualization capabilities for analytics dashboard.	Enhanced analytics dashboard visualizations.	10 hrs	Hard	Mapping natural language queries to appropriate visualization types.	Validation of visualization requirements.	Completed	Learned intent-based visualization generation and KPI reporting concepts.
14-06-2026	Shiva Hariny M R	Memory & Response Quality Improvements	Improve answer quality and version management.	Improved answer quality and implemented version management strategy.	6 hrs	Medium	Maintaining conversation context while ensuring accurate responses	Mentor feedback on response quality.	Completed	Learned conversational memory management, version consistency and response refinement techniques.
15-06-2026	Shiva Hariny M R	Code Modularization	Refactor the existing chatbot into modular components as instructed by mentor.	Refactored chatbot into modular components as per mentor guidance.	7 hrs	Hard	Refactoring existing code without affecting existing functionality.	Code review after modularization.	Completed	Learned modular software architecture, separation of concerns and designing reusable chatbot components.

For Ingate:



IT Consultanting

Date: July 8, 2026

Subject: Internship Completion Letter

TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Shiva Hariny M R** has successfully completed a 30-day internship at **Inavate** from **15/5/2026 to 15/6/2026**

During this tenure, she functioned in the capacity of a **Developer Intern**, primarily contributing to core digital infrastructure initiatives for **Velantri Labs**. Her functional responsibilities encompassed:

- **Full-Stack Application Development:** Assisting with backend and frontend development lifecycles to translate digital product requirements into clean, modular code.
- **AI & Retrieval Engineering:** Designing robust RAG pipelines and retrieval mechanisms to optimize context-driven data translation and system outputs.
- **Integrations, Observability & Guardrails:** Linking APIs and database environments while validating code execution using frameworks like Langfuse to ensure performance and system safety.

Throughout the internship, Shiva Hariny M R demonstrated exceptional dedication, technical aptitude, and a strong work ethic. She proved to be an agile learner capable of adapting to complex systems quickly.

We appreciate her contributions to the team and wish her the very best in all her future academic and professional endeavors.

Best regards,

A blue ink handwritten signature, appearing to read 'Vithya Ragavee R S', written over the 'Best regards,' text.

Vithya Ragavee R S

ACKNOWLEDGEMENT

I first of all love to record my deepest gratitude to the Almighty, my parents and my teachers who brought me up to this stage.

I take immense pleasure for thanking **Rev. Fr. Dr. Godwin Rufus S.J.**, our Principal, **Dr. A. Lourdusamy**, our Deputy Principal for having permitted me to carry out this internship work.

No one grows tired of being to thank, **Dr. A. Natarajan**, our Head of Department who has been a source of inspiration and all time motivator, which played a major role in the conduct of my internship work.

I wish to express my deep sense of gratitude to my Guide, **Dr. Y. Sunil Raj, M.Sc., M.Phil., Ph.D.**, for the guidance and useful suggestions, which helped me in completing the internship work on time.

My Special thanks to my **Mentors** at **Inavate** for their valuable help and encouragement throughout this project. And I thank to all my friends who supported with real encouragement and for their suggestion in each and every phase of this internship.

Shiva Hariny M R

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ABSTRACT

Artificial Intelligence has become a key technology in modern enterprise applications, enabling organizations to efficiently manage business information and support intelligent decision-making. Retrieval-Augmented Generation (RAG) enhances the capabilities of Large Language Models by combining them with enterprise knowledge sources to generate accurate, context-aware, and reliable responses. The primary objective of this internship was to contribute to the development and enhancement of **WorkLyft**, an enterprise AI-powered chatbot that integrates structured CRM data, uploaded business documents, vector search, and external knowledge sources to answer business queries effectively.

During the internship, several enhancements were implemented across the platform, including hybrid retrieval, business KPI calculation, document-to-database comparison, version management, confidence scoring, source attribution, visualization modules, reporting features, role-based authentication, and modular chatbot architecture. The project was developed using technologies such as **Python, FastAPI, PostgreSQL, pgvector, LangChain, Langfuse, React, and JavaScript**, with a strong emphasis on scalable architecture, modular design, testing, debugging, and production-ready software development.

The internship provided valuable exposure to enterprise software engineering practices, AI system design, backend and frontend development, database management, API development, and collaborative software development. It strengthened practical knowledge in Retrieval-Augmented Generation, enterprise AI applications, software modularization, and scalable platform development while bridging the gap between academic learning and real-world industrial experience.

CHAPTER 1

INTRODUCTION TO THE ORGANIZATION & INTERNSHIP

1.1 Introduction

Industrial internships play a vital role in bridging the gap between academic learning and professional practice by providing students with hands-on experience in real-world software development environments. They enable students to apply theoretical concepts, understand industry-standard development practices, and gain practical exposure to modern technologies used in enterprise applications.

As part of the Master of Science in Data Science program, an internship was undertaken at **Inavate Tech**, where practical experience was gained in developing and enhancing **WorkLyft**, an enterprise AI-powered chatbot platform. The project leveraged technologies such as Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), PostgreSQL, FastAPI, React, and vector databases to provide intelligent, context-aware responses using structured CRM data, uploaded business documents, and external knowledge sources.

During the internship, several enhancements were implemented to improve the platform's functionality, including business KPI calculations, hybrid retrieval, document-to-database comparison, version management, confidence scoring, visualization, reporting, role-based authentication, and modular architecture. The internship also provided valuable experience in backend and frontend development, database management, API development, testing, debugging, and collaborative software engineering, thereby strengthening technical skills and providing practical insight into the design and maintenance of enterprise AI applications.

1.2 Company Profile

Inavate Tech is a technology-driven software development company specializing in enterprise solutions powered by AI, ML, cloud technologies, and modern software engineering practices. The company develops software platforms that help organizations automate business processes, manage enterprise knowledge, improve operational efficiency, and support data-driven decision-making.

During the internship, valuable exposure was gained to professional software development practices, including modular software architecture, backend and frontend development, API design, database management, Retrieval-Augmented Generation (RAG), testing, debugging, and collaborative development using Git. Working in a team-based development environment provided practical experience in feature implementation, code integration, version control, and enterprise application development.

1.3 Internship Overview

- Developed and enhanced the **WorkLyft** enterprise AI chatbot using Retrieval-Augmented Generation (RAG).
- Improved chatbot performance by implementing hybrid retrieval, business analytics, visualization, and reporting features.
- Built solutions for document comparison, version management, confidence scoring, and modular architecture.
- Worked with technologies including **Python, FastAPI, PostgreSQL, pgvector, LangChain, React, JavaScript, and Git**.
- Gained practical experience in enterprise software development, debugging, testing, API development, and collaborative coding.

1.4 About the Project – WorkLyft

WorkLyft is an enterprise AI-powered knowledge management and business intelligence platform that enables organizations to interact with their enterprise knowledge through natural language conversations. The platform uses Retrieval-Augmented Generation (RAG) to combine the reasoning capabilities of LLMs with organizational knowledge stored across multiple enterprise data sources.

The system retrieves information from structured CRM databases, uploaded business documents, vector databases, and external information sources before generating responses. This retrieval-first approach improves factual accuracy while reducing hallucinations commonly associated with standalone language models.

The platform supports several enterprise capabilities including intelligent question answering, CRM analytics, business KPI calculation, visualization, report creation, document summarization, uploaded file analysis, internal database querying, external information retrieval, hybrid search, confidence scoring, source attribution, role-based authentication, and enterprise security mechanisms.

A major focus during development was improving the platform's scalability through modular software architecture. Independent service modules were designed to separate retrieval, reporting, visualization, analytics, and chatbot functionalities, making the platform easier to maintain, extend, and integrate with future enterprise applications.

The project demonstrates how Artificial Intelligence can be integrated into enterprise environments to improve knowledge accessibility, automate business analysis, and assist users in making data-driven decisions through conversational interfaces.

1.5 Objectives of the Internship

The major objectives of the internship were:

- To understand enterprise software development methodologies and collaborative programming practices.
- To gain practical experience in Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), and enterprise AI application development.
- To develop and enhance AI chatbot features, including retrieval, business analytics, visualization, reporting, and KPI calculation modules.
- To learn modular software architecture, scalable application design, testing, debugging, and performance optimization techniques.
- To gain hands-on experience in version control, collaborative development, software deployment workflows, and apply academic knowledge to real-world industrial projects.

1.6 Scope of the Internship

The internship covered multiple areas of AI and enterprise software engineering, including Retrieval-Augmented Generation, semantic search, vector databases, REST API development, backend services, frontend integration, enterprise analytics, visualization systems, reporting engines, authentication mechanisms, database management, modular software architecture, and production-level software maintenance.

The knowledge and experience gained during the internship provide a strong foundation for future work in AI, ML, Enterprise Software Development, Backend Engineering, Full Stack Development, and AI Platform Engineering.

CHAPTER 2

LITERATURE SURVEY AND BACKGROUND STUDY

2.1 Introduction

The rapid growth of Artificial Intelligence has significantly transformed the way organizations access, process, and utilize information. Traditional information retrieval systems depend heavily on keyword-based searches and manually organized databases, making it difficult for users to obtain accurate information efficiently. The emergence of Large Language Models (LLMs) has introduced conversational interfaces capable of understanding natural language and generating human-like responses. However, standalone language models often suffer from hallucination, outdated knowledge, and limited access to organization-specific information.

To overcome these limitations, Retrieval-Augmented Generation (RAG) has emerged as one of the most effective approaches for enterprise AI systems. RAG combines document retrieval techniques with language generation, allowing AI models to answer questions using real organizational data instead of relying solely on pre-trained knowledge.

The WorkLyft project utilizes this modern AI architecture by integrating enterprise databases, uploaded documents, vector search, external information retrieval, and business analytics into a unified conversational platform. This chapter provides the theoretical background required to understand the technologies used throughout the internship.

2.2 Artificial Intelligence

Artificial Intelligence (AI) refers to the simulation of human intelligence by computer systems. AI enables machines to perform tasks that traditionally require human reasoning, learning, problem-solving, decision-making, and language understanding.

Modern AI systems utilize Machine Learning and Deep Learning techniques to analyze large datasets, recognize patterns, make predictions, and automate complex business operations.

Applications of Artificial Intelligence include:

- Virtual assistants and Chatbots
- Recommendation systems
- Fraud detection
- Medical diagnosis
- Business analytics
- Predictive maintenance
- Autonomous vehicles

In enterprise environments, AI helps organizations automate repetitive tasks, improve customer support, enhance decision-making, and increase operational efficiency.

2.3 Large Language Models (LLMs)

Large Language Models are deep learning models trained on massive collections of text data to understand and generate natural language. Popular LLMs include: GPT, Claude, Gemini, LLaMA, and Mistral

These models can perform numerous language tasks including:

- Question answering
- Text summarization
- Code generation
- Translation
- Content generation
- Information extraction

Although LLMs possess extensive general knowledge, they cannot automatically access organization-specific information or newly uploaded documents. As a result, they may generate incorrect or outdated responses, commonly referred to as hallucinations.

This limitation motivated the development of Retrieval-Augmented Generation systems.

2.4 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is an AI architecture that combines document retrieval with language generation to produce more accurate and context-aware responses.

Instead of answering solely from the model's internal knowledge, the system first retrieves relevant information from trusted data sources and then provides answers based on those retrieved documents.

The typical RAG workflow consists of:

1. User submits a question.
2. The question is converted into vector embeddings.
3. Similar documents are retrieved from the vector database.
4. Relevant context is passed to the language model.
5. The model generates an answer grounded in retrieved information.

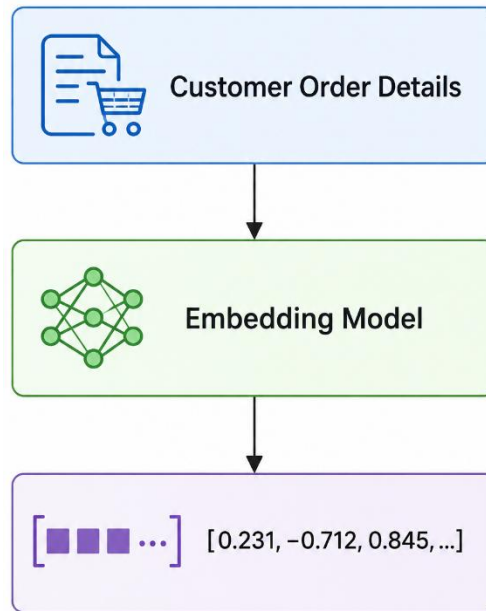
The WorkLyft platform extensively utilizes the RAG architecture to answer business-related queries using CRM records, uploaded documents, and external knowledge sources.

2.5 Vector Embeddings

Vector embeddings are numerical representations of textual information generated by embedding models. Instead of storing text as plain words, embeddings convert text into high-dimensional vectors that capture semantic meaning.

Texts with similar meanings generate vectors located close together within the vector space. **Benefits include:** Semantic search, Similarity matching, Document retrieval, Duplicate detection and Recommendation systems

The WorkLyft platform stores document embeddings inside PostgreSQL using the pgvector extension for efficient similarity search.



2.6 Vector Databases

Vector databases are specialized databases designed to store and retrieve vector embeddings efficiently. Unlike traditional SQL searches that rely on exact keyword matching, vector databases support semantic similarity search.

Popular vector databases include: Pinecone, Weaviate, Milvus, ChromaDB and PostgreSQL pgvector.

The WorkLyft project uses PostgreSQL with the pgvector extension because it allows both structured relational data and vector embeddings to coexist within a single database.

Advantages include:

- Fast semantic retrieval
- Efficient nearest-neighbor search
- Enterprise scalability
- Easy integration with existing SQL databases

2.7 Enterprise AI Chatbots

Enterprise chatbots differ significantly from general-purpose conversational AI systems. General chatbots answer questions based on internet-scale knowledge.

Enterprise chatbots retrieve information from organizational resources such as: CRM databases, internal reports, Uploaded documents, Meeting notes, Business analytics and Knowledge bases

Major characteristics include:

- Secure data access
- Role-based permissions
- Source attribution
- High factual accuracy
- Business-specific responses

The WorkLyft chatbot was designed specifically for enterprise use, supporting CRM retrieval, document analysis, KPI calculations, visualization, and external information retrieval.

2.8 Hybrid Information Retrieval

For example:

Compare today's gold rate with our inventory purchase price.

This requires:

Internal Source: Inventory purchase details

External Source: Live gold market prices

Benefits include:

- Better business insights
- Real-time analysis
- Accurate comparisons
- Improved decision support

During the internship, enhancements were implemented to clearly distinguish between Internal, External, and Hybrid retrieval sources while calculating confidence scores.

2.9 FastAPI Framework

FastAPI is a modern Python framework used for building high-performance REST APIs.

Major features include:

- Automatic API documentation
- High performance
- Asynchronous programming
- Request validation
- Easy integration with AI services

In the WorkLyft project, FastAPI serves as the backend framework responsible for: Authentication, Chat APIs, File uploads, Visualization generation, Business reporting and Retrieval pipeline execution

2.10 PostgreSQL Database

PostgreSQL is an advanced open-source relational database management system widely used in enterprise software.

Features include: ACID compliance, Strong security, SQL support, JSON storage, Extension support and High reliability

The WorkLyft platform stores:

- Customers
- Leads
- Orders
- Marketing strategies
- Uploaded files
- Version history
- Vector embeddings
- User accounts

PostgreSQL forms the central repository for all enterprise business information.

2.11 Data Visualization

Users often understand information better through graphs than raw data. **Visualization techniques include:** Bar charts, Pie charts, Line graphs, KPI cards, Trend analysis and Dashboard reports.

The visualization engine developed during the internship converts retrieved business data into structured visual formats, enabling users to quickly analyze trends and business performance.

2.12 Related Technologies Used

The project integrates several modern technologies:

Technology	Purpose
Python	Backend development
FastAPI	REST API framework
PostgreSQL	Relational database
pgvector	Vector similarity search
React.js	Frontend development
JavaScript	Client-side scripting
HTML & CSS	User interface
Git & GitHub	Version control
JWT	Authentication
REST APIs	Communication between frontend and backend
Large Language Models	Natural language generation
Retrieval-Augmented Generation	Enterprise knowledge retrieval

CHAPTER 3

SYSTEM ANALYSIS AND PROJECT ARCHITECTURE

3.1 Introduction

Enterprise organizations generate large volumes of structured and unstructured data through CRM systems, business reports, meeting notes, sales records, and other operational documents. Traditional search systems often rely on keyword matching, making it difficult to retrieve relevant information efficiently.

The WorkLyft project addresses these challenges by integrating Artificial Intelligence, Retrieval-Augmented Generation (RAG), vector search, structured databases, uploaded documents, and external knowledge sources into a unified conversational platform that delivers accurate, context-aware, and explainable responses to business users.

3.2 Existing System

Before implementing the enhanced WorkLyft platform, several limitations were observed in traditional enterprise chatbot systems.

Most existing enterprise chatbots provide responses based only on predefined FAQs or static knowledge bases. These systems lack the capability to retrieve information dynamically from multiple business sources.

Some common limitations include:

- Inability to retrieve information from structured CRM databases.
- Lack of support for uploaded business documents.

- No semantic search capability.
- Dependence on exact keyword matching.
- Limited ability to answer analytical business questions.
- No version management for uploaded documents.
- No comparison between document data and live CRM data.
- No support for business KPI calculations.
- Limited visualization capabilities.
- Poor explanation of response sources.
- Lack of confidence estimation.
- Difficulty integrating with other enterprise applications.

These limitations reduce the usefulness of enterprise chatbots in real-world business environments where information constantly changes.

3.3 Problem Statement

Organizations require an intelligent conversational platform capable of answering business questions using information from multiple enterprise data sources while maintaining high accuracy, transparency, and scalability.

The system should be capable of:

- Retrieving information from CRM databases.
- Searching uploaded business documents.
- Performing semantic search using vector embeddings.
- Combining internal and external information.
- Supporting business analytics.
- Performing KPI calculations.
- Comparing uploaded reports with live business data.

- Managing multiple versions of uploaded documents.
- Generating business visualizations.
- Providing source attribution.
- Maintaining high retrieval confidence.
- Supporting modular integration into different enterprise platforms.

Developing such a platform requires combining modern AI techniques with scalable software engineering principles.

3.4 Proposed System

The proposed system is an enterprise AI assistant built using Retrieval-Augmented Generation (RAG) architecture. Instead of depending solely on a language model, the chatbot retrieves relevant business information from multiple enterprise sources before generating responses.

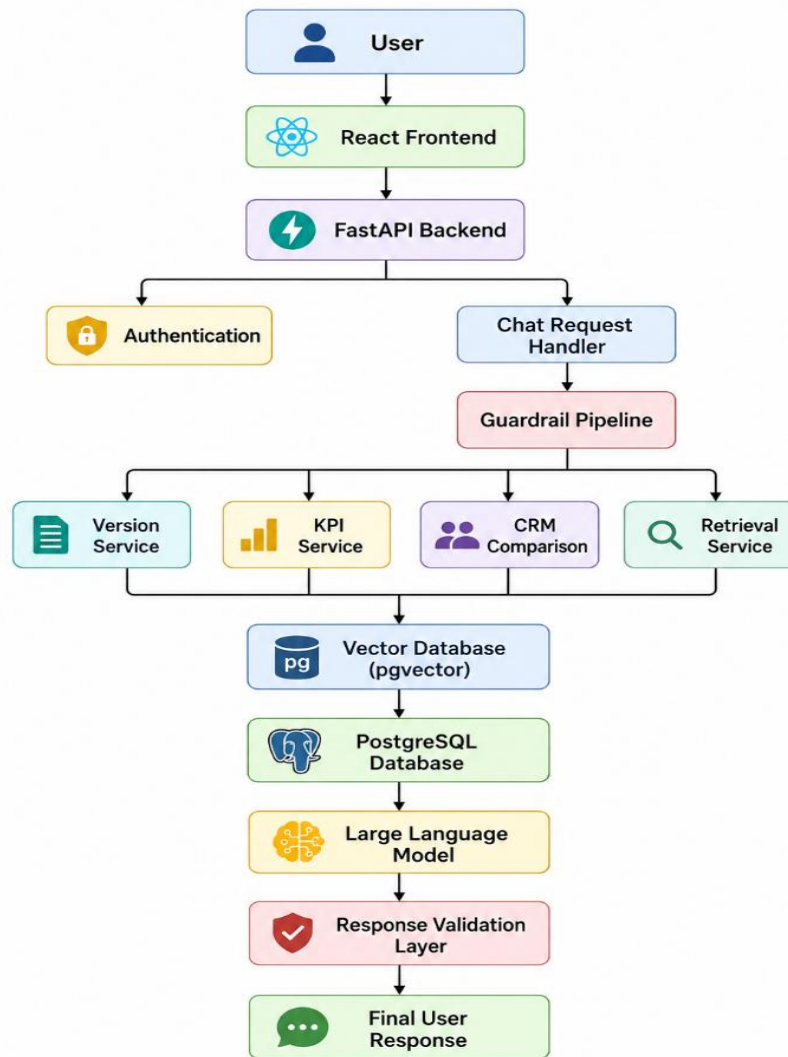
The proposed system integrates:

- Structured CRM database
- Uploaded documents
- Vector similarity search
- External information retrieval
- Reporting engine
- Business analytics
- Visualization engine
- Version-aware document retrieval
- KPI calculation engine
- Document versus CRM comparison
- Confidence scoring and Source attribution

This architecture significantly improves answer accuracy while providing transparent explanations of how answers are generated.

3.5 System Architecture

The WorkLyft platform follows a layered architecture where each module performs a specific responsibility.



The modular architecture allows each component to operate independently, improving maintainability and scalability.

3.6 Major Modules

The WorkLyft platform consists of several interconnected modules.

3.6.1 Authentication Module

Responsible for:

- User login
- JWT authentication
- Role-based access control
- Session validation

3.6.2 Retrieval Module

Responsible for retrieving information from:

- CRM database
- Uploaded files
- Vector embeddings
- External sources

3.6.3 Version Management Module

Features include:

- Version history
- Previous versions
- Latest version retrieval
- Version comparison
- Upload timestamps

3.6.4 KPI Calculation Module

Supported calculations include:

- Lead conversion rate
- Order conversion rate
- Win rate
- Average deal size
- Average order value
- Revenue from won deals
- Order value growth
- Strategy budget analysis
- Total revenue
- Lead loss percentage

3.6.5 Document vs CRM Comparison Module

The module performs:

- Field extraction
- Field normalization
- CRM aggregation
- Difference calculation
- Matching field detection
- Missing field identification

3.6.6 Reporting Module

Generates structured responses for:

- Lead summaries

- Order summaries
- Customer summaries
- Marketing strategies

3.6.7 Visualization Module

Generates graphical representations including:

- Bar charts
- Pie charts
- Line charts
- KPI cards

3.6.8 External Knowledge Module

Examples include:

- Gold prices
- Technology news
- Security vulnerabilities
- Cloud platform updates

3.6.9 Confidence Scoring Module

Each generated response includes:

- Trust score
- Relevance score
- Recency score
- Entity matching score
- Source type

Source classification includes:

- Internal
- External
- Hybrid

3.7 Database Design

Major tables include:

- Customer
- Leads
- Orders
- Marketing Strategy
- File Registry
- Vector Chunks
- Users
- Audit Logs

3.8 Technologies Used

Technology	Purpose
Python	Backend development
FastAPI	REST API framework
PostgreSQL	Relational database
pgvector	Semantic search
React.js	Frontend
JWT	Authentication

Git	Version control
LangChain	AI orchestration
Large Language Models	Response generation
REST APIs	Client-server communication

3.9 Advantages of the Proposed System

The enhanced WorkLyft platform offers several advantages over traditional enterprise chatbot systems.

- Supports natural language interaction.
- Retrieves information from multiple enterprise sources.
- Reduces AI hallucinations using Retrieval-Augmented Generation.
- Supports uploaded documents and version management.
- Performs business KPI calculations.
- Compares uploaded reports with live CRM data.
- Provides visualization support.
- Distinguishes between Internal, External, and Hybrid sources.
- Includes confidence scoring for transparency.
- Modular architecture improves scalability and future integration.

CHAPTER 4

IMPLEMENTATION AND DEVELOPMENT

4.1 Introduction

This chapter presents the implementation work carried out during the internship to enhance the WorkLyft Enterprise AI Chatbot. The development focused on improving retrieval accuracy, scalability, modularity, business analytics, and overall system performance by enhancing the Retrieval-Augmented Generation (RAG) pipeline, implementing document version management, KPI calculation, document-to-database comparison, confidence scoring, guardrails, visualization modules, and modular services.

A modular development approach was followed to ensure that new features were integrated without affecting existing functionalities, and comprehensive regression testing was performed to maintain system stability and reliability.

4.2 Development Environment

The project was developed using the following software and tools.

Software / Tool	Purpose
Python	Backend Development
FastAPI	REST API Framework
PostgreSQL	Database
pgvector	Vector Database Extension

React.js	Frontend Development
Visual Studio Code	IDE
Git & GitHub	Version Control
Postman	API Testing
Pytest	Unit Testing
JWT Authentication	Secure Login
LangChain	RAG Pipeline
Claude / LLM APIs	Response Generation

4.3 Retrieval-Augmented Generation Pipeline

One of the major components of the project is the RAG pipeline. Instead of directly asking the LLM to answer every question, the system first retrieves the most relevant information from enterprise DB, uploaded files, or trusted external sources.

The pipeline consists of the following stages:

1. User Question
2. Question Classification
3. Guardrail Validation
4. Retrieval Layer
5. Vector Similarity Search
6. Database Retrieval
7. External Retrieval (if required)
8. Prompt Construction

9. Language Model Generation

10. Response Validation

11. Confidence Scoring

12. Final Response

This architecture significantly improves answer accuracy while reducing hallucinations.

4.4 Guardrail Pipeline

To ensure reliable responses, a multi-layer guardrail pipeline was implemented.

The guardrail system performs:

- Query validation
- Domain detection
- Entity extraction
- Retrieval quality checking
- Confidence validation
- Response verification

The pipeline prevents the chatbot from answering unsupported or unrelated questions while maintaining factual consistency.

Benefits include: Reduced hallucinations, Higher retrieval precision, Improved security and Better enterprise reliability

4.5 Version Management System

One of the major features developed during the internship was version-aware document retrieval.

Whenever a document is uploaded, it is registered inside the File Registry with version information.

The system automatically maintains:

- Version numbers
- Previous version links
- Upload timestamps
- Latest document identification

The chatbot now supports questions such as:

- Show version history
- Show version 1
- Show latest version
- Compare version 1 and version 3
- How many versions exist?
- When was version 2 uploaded?

Unlike normal retrieval, version-specific queries bypass semantic search and directly retrieve information from the version registry.

Advantages include: Historical document access, Document evolution tracking, Version comparison and better audit capability

4.6 KPI Calculation Engine

A dedicated KPI calculation service was developed to answer analytical business questions directly from CRM data.

Instead of relying on language model reasoning, the system performs SQL aggregation followed by mathematical calculations.

The supported metrics include:

- Lead Conversion Rate
- Order Conversion Rate
- Win Rate
- Lead Loss Percentage
- Average Deal Size
- Average Order Value
- Total Revenue
- Revenue from Won Deals
- Order Value Growth
- Strategy Budget Change

Each response includes:

- Formula used
- Input values
- Calculation steps
- Final result

This approach guarantees deterministic and accurate business analytics.

4.7 Document versus CRM Comparison

Another important enhancement implemented during the internship was document-to-database comparison.

Previously, uploaded documents could only be searched independently.

The new comparison module allows the chatbot to compare uploaded reports against live CRM information.

The comparison process consists of:

1. Detect comparison intent.
2. Identify uploaded document.
3. Extract structured key-value pairs.
4. Retrieve current CRM statistics.
5. Normalize field names.
6. Match similar fields.
7. Calculate differences.
8. Generate comparison report.

The comparison identifies:

- Matching fields
- Changed values
- Missing values
- Additional CRM information
- Additional document information

This feature allows business users to compare reports with current operational data.

4.8 Hybrid Retrieval Enhancement

The chatbot retrieves information from multiple sources.

Three retrieval modes are supported.

1. Internal Retrieval

Uses only CRM database and uploaded enterprise documents.

Example:

Which leads are won?

2. External Retrieval

Uses only trusted external sources.

Example:

What is today's gold rate?

The confidence panel correctly displays: **Source: External**

3. Hybrid Retrieval

Combines internal enterprise data with external information.

Example:

Compare today's gold rate with our inventory purchase price.

The chatbot retrieves:

Internal:

- Inventory purchase price

External:

- Live market gold rate

The confidence panel displays: **Source: Hybrid**

This enhancement improves transparency and clearly indicates how responses are generated.

4.9 Visualization Improvements

The visualization engine was enhanced to improve business analytics.

The module now supports:

- Better chart validation
- Data normalization
- KPI cards
- Chart configuration
- Correlation analysis
- Outlier detection
- Improved dashboard generation

Additional validation ensures that invalid datasets do not generate misleading visualizations.

4.10 Confidence Scoring

Every chatbot response includes a confidence assessment.

The confidence score is calculated using multiple factors:

- Trust
- Relevance
- Recency
- Entity Match

The system also indicates whether the response originates from:

- Internal Sources
- External Sources
- Hybrid Sources

This helps users understand the reliability of generated responses.

4.11 Source Attribution

Every generated answer includes source references.

Possible sources include:

- CRM Database
- Uploaded Documents
- External Websites
- Hybrid Retrieval

Displaying sources improves transparency and allows users to verify the information used to generate responses.

4.12 Modular Architecture Enhancement

Towards the end of the internship, the chatbot architecture was partially modularized to prepare it for future integration into multiple platforms.

The implementation focused on separating the core chatbot functionality from the application-specific components.

The architecture was reorganized into independent services responsible for:

- Chat Engine
- Retrieval Service
- Reporting Service
- Version Service
- KPI Service
- Visualization Service
- Authentication Service

The orchestration layer was updated to coordinate communication between these services while preserving the existing Retrieval-Augmented Generation workflow.

This modular structure improves maintainability, enables easier testing, and prepares the chatbot for future deployment through REST APIs, SDKs, web widgets, or Model Context Protocol (MCP) interfaces. During the internship, the **core modularization work** was completed, while advanced integration features were identified as future enhancements.

4.13 Testing and Validation

After implementing every enhancement, extensive regression testing was performed.

Testing included:

- Unit Testing

- Integration Testing
- Regression Testing
- API Testing
- Version Retrieval Testing
- KPI Calculation Testing
- Document Comparison Testing
- Visualization Testing

More than **500 automated test cases** were executed successfully during development to verify that new features did not affect existing functionalities.

4.14 Challenges Faced

Several technical challenges were encountered during development.

Some major challenges included:

- Preventing regressions while adding new features.
- Designing generic solutions without hardcoding business entities.
- Maintaining compatibility with existing retrieval pipelines.
- Implementing accurate business metric calculations.
- Comparing heterogeneous document and CRM data.
- Preserving retrieval performance while introducing version-aware queries.
- Resolving merge conflicts while integrating parallel development work.
- Maintaining modularity without disrupting existing application behavior.

Each challenge was addressed through careful architectural planning, modular development, regression testing, and iterative improvements.

CHAPTER 5

RESULTS, TESTING AND PERFORMANCE EVALUATION

5.1 Introduction

This chapter presents the testing procedures, experimental results, and performance evaluation of the WorkLyft Enterprise AI Chatbot after implementing the enhancements developed during the internship. The primary objective of testing was to verify that all newly implemented features function correctly while ensuring that previously existing functionalities remained unaffected.

Each feature was tested using multiple real-world business queries covering CRM retrieval, uploaded document analysis, version-aware retrieval, KPI calculation, hybrid information retrieval, document comparison, visualization, and external knowledge retrieval. Regression testing was also performed to confirm that newly introduced modules did not introduce errors into the existing system.

The evaluation demonstrated that the chatbot successfully handled enterprise-level business queries with high retrieval accuracy while maintaining explainable responses through confidence scoring and source attribution.

5.2 Testing Methodology

A systematic testing approach was followed throughout the internship. Every newly implemented module was validated independently before performing full system regression testing.

The testing process included:

- Unit Testing
- Integration Testing
- Functional Testing
- Regression Testing
- API Testing
- Manual Validation
- User Acceptance Testing
- Performance Verification

Both positive and negative test cases were executed to ensure that the chatbot behaved correctly under different scenarios.

5.3 Testing Environment

The application was tested using the following environment.

Component	Details
Operating System	Windows 11
Backend	FastAPI
Frontend	React.js
Database	PostgreSQL with pgvector
Programming Language	Python
API Testing	Postman
Unit Testing	Pytest
Browser	Google Chrome
Version Control	GitHub

5.4 CRM Retrieval Testing

The chatbot was evaluated using multiple CRM-related business questions.

Sample test cases included:

- List all leads.
- Which leads are won?
- Which leads are lost?
- Which leads are in closure stage?
- Show all orders.
- Show all marketing strategies.
- Who is Arjun Mehta?
- Tell me about Quantum Flow.

The chatbot successfully retrieved information directly from the CRM database without requiring language model inference.

Observed outcome:

- Accurate entity matching
- Structured responses
- Correct business statistics
- High confidence scores

5.5 Uploaded Document Retrieval Testing

Uploaded business documents were tested to verify semantic retrieval.

Example questions included:

- What are the agreed next steps?
- Who conducted the discovery call?

- What are the key pain points?
- Which competitor was mentioned?
- Tell me about Raja Bullion Traders.

Results showed that the chatbot successfully extracted relevant information from uploaded files while maintaining proper source attribution.

5.6 Version Management Testing

The version-aware retrieval module was extensively tested.

Supported queries included:

- Show version history.
- Show version 1.
- Show version 2.
- Show latest version.
- Show previous version.
- How many versions are available?
- When was version 2 uploaded?

The chatbot correctly retrieved historical document versions without affecting normal retrieval operations.

The version management feature also displayed upload timestamps and complete version history from the file registry.

5.7 KPI Calculation Testing

The KPI calculation engine was validated using several analytical business questions.

Examples included:

- What is the lead conversion rate?
- What is the order conversion rate?
- What is the win rate?
- What is the average deal size?
- What is the average order value?
- What is the total revenue?
- What is the lost lead percentage?
- What is the strategy budget change?
- What is the order value growth?

Unlike language-model-generated answers, all KPI calculations were produced directly from SQL aggregation queries using predefined mathematical formulas.

Each response contained:

- Formula used
- Input values
- Intermediate calculations
- Final result

This ensured deterministic and reproducible business analytics.

5.8 Hybrid Retrieval Testing

One important enhancement involved distinguishing between Internal, External, and Hybrid retrieval.

Three categories of questions were tested.

Internal Retrieval

Example: Which leads are won?

Result: Source displayed as: **Internal**

External Retrieval

Example: What is today's gold rate?

Result: Source displayed as: **External**

Hybrid Retrieval

Example: Compare today's gold rate with our inventory purchase price.

Result: The chatbot retrieved:

Internal:

- Inventory purchase price

External:

- Live gold market price

Source displayed as: **Hybrid**

This enhancement improved response transparency and helped users understand how answers were generated.

5.9 Document versus CRM Comparison Testing

A new comparison module was developed to compare uploaded reports against live CRM data.

Example queries included:

- Compare uploaded CRM comparison test with actual orders.
- Compare uploaded CRM comparison test against current leads.
- Compare uploaded CRM comparison test with current strategy spend.

The system successfully identified:

- Matching fields
- Changed values
- Missing fields
- Additional CRM fields
- Additional document fields

Difference calculations were automatically generated for numerical values, providing users with detailed comparison reports.

5.10 Confidence Scoring Evaluation

Every chatbot response included confidence information.

The confidence panel displayed:

- Trust Score
- Relevance Score
- Recency Score
- Entity Matching Score
- Source Type

This provided explainability and improved user trust in generated responses.

5.11 Regression Testing

Regression testing was performed after every enhancement to ensure that existing functionalities remained operational.

The following modules were verified repeatedly:

- CRM Retrieval
- Uploaded File Retrieval
- Reporting Engine
- Version Retrieval
- KPI Calculations
- Visualization
- Hybrid Retrieval
- Authentication
- External Search

No functional regressions were observed after integrating the new modules.

More than **500 automated test cases** executed successfully, confirming the stability of the application.

5.12 Performance Evaluation

The enhanced system demonstrated significant improvements in functionality and usability.

Feature	Result
CRM Retrieval	Successful

Uploaded Document Retrieval	Successful
Version Retrieval	Successful
KPI Calculation	Successful
Document Comparison	Successful
Hybrid Retrieval	Successful
Confidence Scoring	Successful
Source Attribution	Successful
Modular Services	Successful
Regression Testing	Passed

The modular architecture also improved maintainability by separating responsibilities into independent services.

5.13 Advantages Achieved

The implemented enhancements provided several practical benefits.

- More accurate enterprise information retrieval.
- Reliable KPI calculations using SQL.
- Transparent confidence scoring.
- Better distinction between Internal, External, and Hybrid sources.
- Historical document retrieval through version management.
- Automatic comparison of uploaded reports with CRM data.
- Modular architecture supporting future scalability.
- Reduced AI hallucinations through Retrieval-Augmented Generation.

5.14 Limitations

Although the system achieved its intended objectives, certain limitations remain.

- Comparison is currently limited to structured fields extracted from uploaded documents.
- The modular architecture currently focuses on the core chat engine, while integrations such as SDKs, JavaScript widgets, and Model Context Protocol (MCP) support are planned for future development.
- External information retrieval depends on the availability and reliability of trusted external sources.
- Visualization capabilities are currently designed for supported business analytics and can be expanded further for more advanced dashboarding.

These limitations provide opportunities for future enhancements.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Introduction

The internship at **Inavate Tech** provided an excellent opportunity to apply theoretical knowledge acquired during academic studies to real-world enterprise software development. Working on the **WorkLyft Enterprise AI Chatbot** offered practical exposure to modern Artificial Intelligence technologies, Retrieval-Augmented Generation (RAG), backend development, frontend integration, database management, software architecture, and collaborative software engineering practices.

Throughout the internship, multiple enhancements were implemented to improve the chatbot's intelligence, scalability, transparency, and overall performance. The experience significantly strengthened both technical knowledge and problem-solving skills while providing valuable insight into the software development lifecycle followed in an industrial environment.

6.2 Internship Summary

The internship primarily focused on enhancing the WorkLyft Enterprise AI platform by implementing new capabilities and improving existing functionalities.

Several important modules were developed and enhanced during the internship, including:

- Retrieval-Augmented Generation (RAG) improvements
- Enterprise retrieval pipeline optimization

- CRM reporting enhancements
- Uploaded document retrieval
- Version-aware document management
- Business KPI calculation engine
- Document versus CRM comparison module
- Hybrid retrieval improvements
- Confidence scoring enhancements
- Visualization improvements
- Guardrail validation
- Source attribution
- Backend modularization
- Testing and regression validation

Each enhancement was designed to improve the reliability, maintainability, and usability of the enterprise chatbot while ensuring compatibility with the existing system.

6.3 Objectives Achieved

The objectives defined at the beginning of the internship were successfully accomplished.

The internship successfully achieved:

- Practical understanding of Retrieval-Augmented Generation systems.
- Experience in developing enterprise Artificial Intelligence applications.
- Integration of structured databases with Large Language Models.
- Implementation of business analytics modules.
- Development of scalable backend services.

- Enhancement of enterprise information retrieval.
- Improvement of chatbot transparency through confidence scoring.
- Implementation of document version management.
- Development of document comparison capabilities.
- Exposure to collaborative software development using Git and GitHub.
- Understanding of modular software architecture.

These achievements contributed significantly to the overall functionality of the WorkLyft platform.

6.4 Technical Skills Acquired

The internship enhanced practical knowledge in several technical areas.

Programming Languages

- Python
- JavaScript
- SQL

Frameworks

- FastAPI
- React.js

Databases

- PostgreSQL
- pgvector

Artificial Intelligence

- Retrieval-Augmented Generation (RAG)

- Large Language Models
- Semantic Search
- Vector Embeddings

Software Development

- REST API Development
- Authentication
- Backend Architecture
- Frontend Integration
- Version Control using Git
- Testing and Debugging
- Modular Software Design

The internship also improved the ability to understand and work with large enterprise codebases.

6.5 Challenges Faced

Several technical challenges were encountered during the internship. Some of the major challenges included:

- Integration of Multiple Data Sources
- Maintaining Backward Compatibility
- Version Management
- Business Analytics
- Document Comparison
- Modularization

6.6 Learning Outcomes

The internship provided valuable practical experience beyond classroom learning.

Major learning outcomes include:

- Understanding enterprise AI architecture.
- Working with Retrieval-Augmented Generation pipelines.
- Managing enterprise databases.
- Implementing modular software architecture.
- Performing collaborative software development.
- Writing production-quality Python code.
- Debugging complex software systems.
- Conducting regression testing.
- Resolving Git merge conflicts.

These experiences significantly improved both technical competence and professional confidence.

6.7 Industrial Exposure

The internship provided insight into professional software engineering practices followed within an enterprise environment.

Exposure was gained to:

- Requirement analysis
- Feature planning
- Sprint-based development

- Code reviews
- Collaborative programming
- Version control workflows
- Testing strategies
- Bug fixing
- Production issue resolution
- Documentation
- Software maintenance

Understanding these industrial practices has prepared the intern for future software engineering roles.

6.8 Future Scope

Although the WorkLyft platform currently provides extensive enterprise AI capabilities, several enhancements can further improve the system.

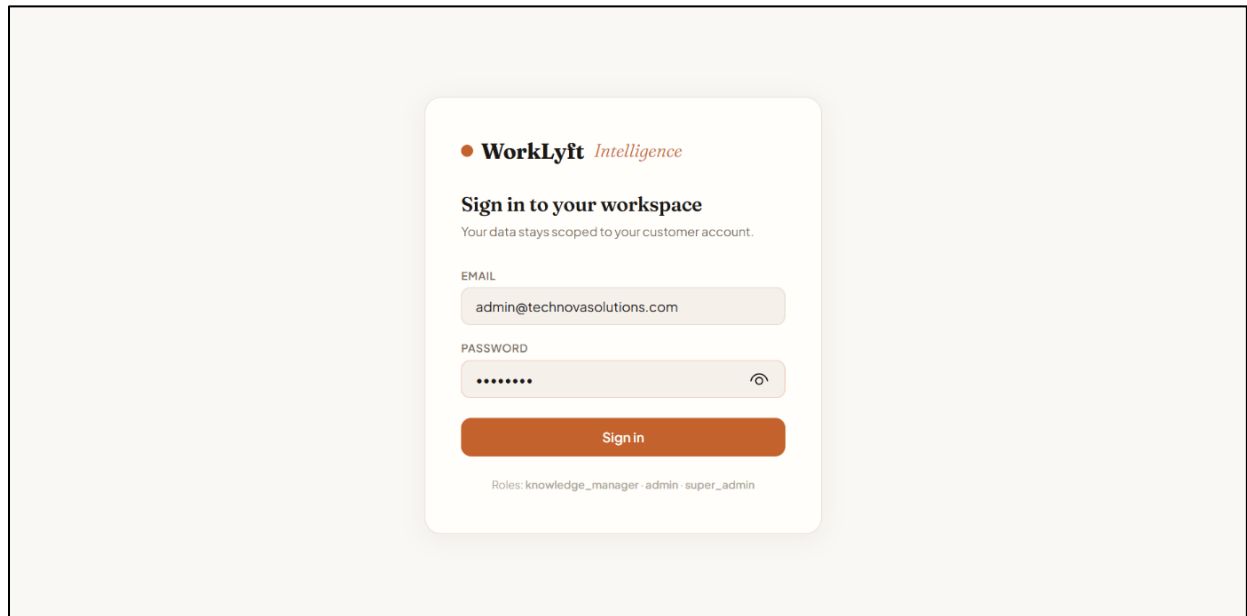
Possible future developments include:

- Complete modularization of all chatbot components into independent services.
- Development of official Python and JavaScript SDKs.
- Integration using the Model Context Protocol (MCP) for AI clients.
- Deployment as an embeddable website widget.
- Support for additional enterprise databases.
- Advanced dashboard analytics.
- Real-time streaming responses.

- Multi-language support.
- Voice-based conversational interface.
- AI-assisted report generation.
- Automated workflow execution based on chatbot responses.
- Improved document comparison using semantic field matching.
- Enhanced visualization dashboards with interactive analytics.
- Multi-tenant deployment for supporting multiple organizations securely.

These enhancements would further strengthen the platform as a scalable enterprise AI solution.

GUI Of WorkLift



The image shows a login page for WorkLift Intelligence. It features a central white card on a light gray background. The card has the WorkLift Intelligence logo at the top, followed by the heading "Sign in to your workspace" and a subtext "Your data stays scoped to your customer account." Below this are input fields for "EMAIL" (containing "admin@technovasolutions.com") and "PASSWORD" (masked with dots). A "Sign in" button is positioned below the password field. At the bottom of the card, it lists roles: "Roles: knowledge_manager · admin · super_admin".

Login Page

Schemas			^
Body_chat_with_file_chat_with_file_post >	Expand all	object	
Body_upload_csv_upload_csv_post >	Expand all	object	
Body_upload_pdf_upload_pdf_post >	Expand all	object	
ChatRequest >	Expand all	object	
ChatResponse >	Expand all	object	
HTTPValidationError >	Expand all	object	
SourceItem >	Expand all	object	
ValidationError >	Expand all	object	

Fastapi Swagger Endpoints

default		^
GET	/health Health	▼
GET	/stats/{customer_id} Get Stats	▼
POST	/chat Chat	▼
POST	/chat-with-file Chat With File	▼
POST	/upload/csv Upload Csv	▼
POST	/upload/pdf Upload Pdf	▼
GET	/threshold Get Threshold	▼
POST	/threshold Update Threshold	▼
GET	/versions/{customer_id} Get Versions	▼

Fastapi Swagger Endpoints

▼	localhost
▼	worklyft
▼	public
>	Functions
>	activity
>	app_user
>	audit_log
>	channel
>	customer
>	file_registry
>	lead
>	marketing_strategy
>	order
>	refresh_token
>	value_chain
>	vector_chunk

Database Structure

which are active

U

W

TechNova Solutions currently has one active strategy, which is the "Q2 2025 Product-Led Growth Push."

2 sources retrieved

100% confidence

which are complete

U

W

In the context provided, the completed strategies for TechNova Solutions are:

- Q1 2025 SaaS Market Expansion
- Q2 2025 Product-Led Growth Push

1 source retrieved

92% confidence

Internal source questioning

who conducted the discovery call

U

W

The discovery call was conducted by Neha Kapoor from TechNova and Arjun Mehta, the founder, who joined for the first 20 minutes.

1 source retrieved

Uploaded file questioning

Show me order value growth

U

W

****Order Value Growth****

First Half Total = Rs 9.35 L
 Second Half Total = Rs 16.40 L
 Total Orders = 6

Formula: $\text{Growth} = ((\text{Second Half} - \text{First Half}) / \text{First Half}) \times 100$

****Result: +75.40%****

_Orders split chronologically by created_at into two equal halves._

1 source retrieved

100% confidence

KPI calculation

what is today's gold rate

U

W

Today's gold rate is Rs 14,908.00 per gram. [Source: www.goodreturns.in, Retrieved: 2026-06-14]

3 sources retrieved

61% confidence

External search

Trust		60%
Relevance		58%
Recency		100%
Entity match		0%

External source retrieval

compare today's gold rate with our inventory purchase details

U

W

Internal value: Rs 5,820 per gram (from: [CHUNK 1])

Current value: Rs 14,908.00 per gram (from: External Live Data)

Difference: $\text{Rs } 14,908.00 - \text{Rs } 5,820 = \text{Rs } 9,088.00$

Change: $((\text{Rs } 14,908.00 - \text{Rs } 5,820) / \text{Rs } 5,820) \times 100 = 157.12\%$ increase

Business Answer: Today's gold rate is significantly higher than our internal purchase price, showing a substantial increase of 157.12%, which could impact profit margins and resale strategies for Raja Bullion Traders.

3 sources retrieved

64% confidence

Hybrid

Trust		60%
Relevance		66%
Recency		100%
Entity match		0%

Hybrid response retrieval

WorkLyft Intelligence Chat Visualize DB mode — answering from pgvector + New chat Platform Admin Super Admin pgvector · GPT-3.5 · Langfuse

CUSTOMER

- Arjun Mehta
TechNova Solutions
- Priya Sharma
Meridian Retail Group
- Rahul Verma
BlueRidge Logistics

KNOWLEDGE BASE

171 Total Chunks	14 Analytics Summary
3 Customer	5 Customer 1 Meeting Notes.Txt
2 Gold Business.Txt	125 Lead
10 Lead Summary	2 Marketing Strategy

Who conducted the discovery call?

Neha Kapoor from TechNova conducted the discovery call with CloudMorph India.

1 source retrieved

92% confidence

Tell me about Raja Bullion Traders

Raja Bullion Traders is a company based in Chennai, Tamil Nadu, specializing in retail and wholesale gold distribution. They deal in products like 24K Gold Bars, 22K Gold Coins, Gold Jewellery, and Investment Bullion. The company monitors daily gold market prices to compare with internal inventory purchase prices for profit margin and resale strategies.

1 source retrieved

95% confidence

Ask anything about TechNova Solutions...

attach file Enter send Shift+Enter new line

Sidebar upload stores permanently to DB

Questioning different uploaded files at the same time

show version history of quarterly business review

****Version History — 'quarterly business review'****

Version 1 → quarterly_business_review_v1.txt
Uploaded: 2026-06-07 00:04:59 (first version)

Version 2 → quarterly_business_review_v2.txt
Uploaded: 2026-06-07 00:05:59 replaces v1

Version 3 → quarterly_business_review_expansion.txt
Uploaded: 2026-06-07 00:06:58 replaces v2

Total: 3 version(s).

3 sources retrieved

98% confidence

Version history retrieval

STORE TO KNOWLEDGE BASE

+

Upload file

PDF CSV TXT

⚠ Duplicate file blocked

×

This content already exists as "contract_v1.txt". Upload a file with different content.

EXACT BYTE MATCH (SHA-256)

Byte-identical file already exists

Duplicate files are detected automatically
 \u00b7 Updated files are versioned automatically.

409
Undocumented

Error: Conflict

Response body

```
{
  "detail": {
    "status": "duplicate",
    "message": "This file has already been uploaded as 'contract_v1.txt'. Upload a file with different content.",
    "conflicting_filename": "contract_v1.txt",
    "nearest_match_similarity": null,
    "duplicate_threshold": 0.95,
    "update_threshold": 0.75,
    "similarity_method": "sha256_hash",
    "classification_reason": "hash_match"
  }
}
```

Response headers

```
access-control-allow-credentials: true
content-length: 336
content-type: application/json
date: Fri,05 Jun 2026 10:36:46 GMT
server: uvicorn
```

Duplicate detection

Response body

```
{
  "status": "updated",
  "conflicting_filename": null,
  "filename": "budget_april.txt",
  "chunks_parsed": 1,
  "chunks_stored": 1,
  "updates_found": 0,
  "customer_id": 2,
  "similarity_score": 0.8203,
  "sample": "Budget: 6 Lakhs Users: 200 Location:Banglore"
}
```

Download

Response headers

```
access-control-allow-credentials: true
content-length: 230
content-type: application/json
date: Fri,05 Jun 2026 04:37:47 GMT
server: uvicorn
```

File updating

Response body

```
{
  "status": "ingested",
  "conflicting_filename": null,
  "filename": "budget_march.txt",
  "chunks_parsed": 1,
  "chunks_stored": 1,
  "updates_found": 0,
  "customer_id": 2,
  "similarity_score": 0.3293,
  "sample": "Budget: 5 Lakhs Users: 100 Location: Chennai"
}
```

Download

Response headers

```
access-control-allow-credentials: true
content-length: 231
content-type: application/json
date: Fri,05 Jun 2026 04:35:51 GMT
server: uvicorn
```

File ingestion

what are the strategies of technova solutions?

U

W

The strategies of TechNova Solutions are:

1. Q1 2025 SaaS Market Expansion (Budget: ₹450,000.00, Status: completed, Period: 2025-01-01 to 2025-03-31)
2. Q2 2025 Product-Led Growth Push (Budget: ₹320,000.00, Status: active, Period: 2025-04-01 to 2025-06-30)

1 source retrieved

lead_summary #900006

100% confidence

Trust	100%
Relevance	100%
Recency	100%
Entity match	100%
Source: internal db	

Source retrieval and confidence scoring

WorkLyft Intelligence

DB mode — answering from pgvector

+ New chat pgvector · GPT-3.5 · Langfuse

CUSTOMER

Arjun Mehta
TechNova Solutions

Priya Sharma
Meridian Retail Group

Rahul Verma
BlueRidge Logistics

KNOWLEDGE BASE

157 Total Chunks	1 AI SaaS Customer Text
3 Analytics Summary	3 Customer
2 Gold Business Text	125 Lead
10 Lead Summary	2 Marketing Strategy

Ask about *TechNova Solutions*

Type a question to search the database, or attach a file with the button to ask about its contents.

Which leads are in Closure stage?

Show me won deals

What orders are delivered?

Who is the contact at FinEdge?

What is the strategy budget?

Ask anything about TechNova Solutions...

attach file to question · Error · Retry · new chat

Webpage content stored permanently to DB

UI

What was the previous budget and current budget?

U

W

Previous budget: 450000.00
Current budget: 320000.00
Difference: 130000.00
Source of update: CHUNK 2, CHUNK 1

DB mode

1.85s

6 chunks

4 sources retrieved

Version comparison

References

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